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AI-Powered Integrated Project Monitoring and Risk Prediction Platform

Abstract:

Large-scale projects involve multiple activities, resources, financial components, timelines, and stakeholders. Monitoring these parameters manually can make it difficult to identify delays, cost escalation, resource-related issues, and emerging project risks at an early stage. This can lead to delayed corrective actions and inefficient project management.

Our proposed solution is an **AI-powered web-based integrated project monitoring platform** that brings important project information into a unified system. The platform will allow project managers to record and monitor

project progress, planned and actual expenditure, timelines, milestones, resources, and other relevant parameters through a centralized dashboard.

The system will use **data analytics and machine learning techniques** to analyse project information and identify potential risks. It will provide indicators for **schedule delay, cost overrun, and overall project risk**, helping stakeholders understand the current condition of a project and take timely corrective action.

Key Features:

- Centralized project monitoring dashboard
- Project progress and milestone tracking
- Planned vs. actual cost analysis
- Schedule and delay monitoring
- AI/ML-based project risk prediction
- Cost-overrun and delay-risk indicators
- Risk-factor analysis and visualization
- Automated alerts and actionable recommendations
- Project-wise analytics and reports
- AI-assisted project information and insights
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Proposed Workflow:

Project Data → Data Processing → AI/ML Analysis → Risk Prediction → Dashboard → Recommendations

The platform will collect project-related information, process and analyse the data, identify deviations from planned performance, and generate understandable risk indicators. The dashboard will present the results through charts, statistics, alerts, and recommendations.

Innovation:

The proposed system goes beyond conventional project tracking by combining **project monitoring, predictive analytics, risk assessment, and intelligent recommendations** in a single platform. Instead of only displaying the current status of a project, the system aims to identify potential problems at an early stage and provide decision-support information to project stakeholders.

Expected Outcome:

The proposed solution aims to improve project visibility, enable early identification of potential delays and cost escalation, support data-driven decision-making, and reduce the impact of project risks through timely intervention. The prototype will demonstrate how AI-driven monitoring can assist project stakeholders in managing complex projects more efficiently.

Technologies:

Frontend: HTML, CSS, JavaScript
Backend: Python, Flask
AI/ML: Python, Pandas, Scikit-learn
Database: SQLite
Visualization: Chart.js
Version Control: Git and GitHub